

Zen Nguyen

(672) 673-2613 | ttn54@sfu.ca | linkedin.com/in/zennguyen1305 | github.com/ttn54 | zennguyen.me

TECHNICAL SKILLS

- **Languages:** Python, Go, TypeScript, Java, C++, SQL
- **AI & Data:** RAG Architecture, Vector Databases (FAISS/ChromaDB), LangChain, OpenAI API, Anthropic API
- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC, TCP Sockets, Raft Consensus Algorithm
- **Infrastructure & DB:** Docker, PostgreSQL, MongoDB, Git, Linux
- **Testing & Validation:** JUnit, Pytest

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 - Present

Personal Project · Python, Go, LangChain, FAISS

- Architected an autonomous AI Co-Pilot in Python that automates end-to-end job application workflows, enabling the discovery, extraction, and parsing of thousands of Software Engineering postings with advanced anti-bot evasion via concurrent JobSpy scrapers.
- Built a zero-hallucination RAG (Retrieval-Augmented Generation) engine using LangChain, OpenAI APIs, and a sandboxed FAISS Vector Store, delivering uniquely tailored, ATS-optimized PDF resumes for each job by leveraging only verified user data.
- Developed TitanStore, a distributed memory bank in Go, implementing the Raft consensus protocol and a custom Write-Ahead Log (WAL) to guarantee application state durability and fault tolerance across network failures.
- Designed and implemented a high-throughput TCP client API, enabling the Python AI workers to perform concurrent, real-time status updates (e.g., Pending, Submitted) to the Go database cluster, supporting seamless scaling to 100+ users.

SFU Course Tracker (sfucourseplanner.me)

Nov 2025 - Jan 2026

Personal Project · Python, Docker, AWS, FastAPI

- Migrated full-stack platform from Azure to AWS (EC2), implementing Docker container orchestration to optimize deployments and reduce infrastructure costs.
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).
- Designed a normalized schema using SQLAlchemy with JSON-type columns to store recursive tree structures, optimizing read performance for complex queries.

Gridlock Casino (2D Arcade Engine)

Feb 2026 - Present

Collaborative Project · Java, Swing, Maven, JUnit

- Architected a custom 2D grid-based arcade game engine in Java, managing game state, 60-FPS rendering cycles, and concurrent user input within a strict MVC architecture.
- Managed project build automation and dependency resolution using Apache Maven to synchronize a 6-developer team, and validated core engine logic through comprehensive JUnit testing.

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025